pNFS RWX Multi-Client Integrity Tester

Specification

Purpose

Detect lost, stale, torn and misplaced data on a shared (RWX) NFS/pNFS volume. The tool runs long, randomized, concurrent read/write load from many clients, optionally with faults injected, and checks every result against an external ground truth. Protocol conformance is out of scope; NetApp nfstest covers that.

Components

- **Coordinator**: a single process outside the volume under test that holds the ledger. For each block it tracks `written` (highest generation handed out) and `committed` (highest generation known to be flushed). It also stores the run config and collects worker reports. It must stay reachable when NFS traffic is partitioned.
- **Init**: creates F files of B blocks each, of size BS, fills them with generation-0 blocks, fsyncs, and resets the ledger. Files are fully preallocated, so size changes never interfere with data checks.
- **Worker**: one per client node, single process (fcntl locks do not conflict within one process). Runs randomized operations for a set duration.
- **Verifier**: after all workers report, reads every block and checks it against the ledger. Run it from every client node.

Block format

```
magic | file_id | writer_id | block_idx | generation | pseudo-random payload | BLAKE2b digest
```

The payload is derived deterministically from the header. Any reader can classify a block on its own as `ok`, `short`, `zero`, `checksum` (torn or corrupt) or `misplaced` (header does not match its position).

Modes

Mode	Writes	Reads	Invariant
`owned`	Each block has one owner, no locks; fsync with probability p, then commit all pending writes	Reopen the file before each read (close-to-open)	`committed_before ≤ gen ≤ written_after`; writer is the owner
`locked`	Any block, under an exclusive byte-range lock; fsync optional; commit before unlock	Under a shared byte-range lock	`gen == committed`
`racy`	Any block, no coordination	Reopen before each read	Integrity only; `gen ≤ written`; torn blocks counted, or fail with `--strict`

Owner scope in `owned` mode is either `file` (pure close-to-open case) or `block` (writers interleaved within files).

Parameters

- **Init**: files, blocks, block size (aligned and unaligned, e.g. 4096 and 3000), workers, mode, owner scope.
- **Worker**: id (or taken from the Job index), duration, write ratio, fsync probability, `--no-reopen` (stale reads expected; counted instead of failed), seed, dump directory.

Checks and outputs

Category	Items
Violations per read	`short`, `zero`, `checksum`, `misplaced`, `stale_read`, `phantom_gen`, `wrong_writer`, `lock_coherence`
Violations any time	`io_error`, recorded with its errno
Violations, verifier only	`size`, `lost_write`
Informational counters	`newer_than_committed`: unflushed data visible to other clients. `uncommitted_tail`: a writer died between writing and flushing. `stale_cached`: stale reads in `--no-reopen` mode.

- **Event log**: violations as JSON lines, with raw blocks optionally dumped to disk.
- **Worker report**: per-worker JSON summary with node name, seed, the `pnfs=` layout type, and before/after counts of LAYOUTGET, LAYOUTCOMMIT, LAYOUTRETURN, LOCK and other ops from `/proc/self/mountstats`.
- **Exit code**: non-zero if there were any violations.

Deployment

- **Kubernetes layout**: coordinator as a Deployment plus Service; init as a Job; workers as an Indexed Job with pod anti-affinity on hostname. Pods on the same node share the NFS client cache, which would hide coherency bugs.
- **Per-index failures**: set `backoffLimitPerIndex: 0` so one failing worker does not kill the rest.
- **Sequencing**: workers wait until init has finished; the verifier waits for N reports.
- **Fresh runs**: restart the coordinator between runs so the ledger starts empty.

Fault scenarios (driven externally)

- Kill a worker pod: committed writes must survive.
- MDS failover or restart: covers grace-period lock reclaim.
- Data-server failure.
- Partition one client from NFS longer than the lease, let others write, heal the partition, then verify. Committed data must not be overwritten by the late client; this tests fencing on block/SCSI layouts.

Out of scope (possible extensions)

Metadata races (create, rename, unlink), O_APPEND across clients, mmap, O_DIRECT, file growth and LAYOUTCOMMIT size visibility, and throughput measurement (use fio for that).